

**International Governance in the Rio Grande/Bravo Basin:
Mechanisms for Improved Governance in Response to Drought and Development**

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Introduction

The Rio Grande River, called the Rio Bravo in Mexico (hereafter, the RGB), is central to the basin's economic success, with 5-6 million people relying on it for drinking water across the two countries (USIBWC, 2024). The RGB has faced significant resource strain due to severe, historically unprecedented droughts compounded by rapid population growth in the region (USIBWC, 2024; Neva et al., 2016). As a result of hydrological changes and increased human consumption, the legal and institutional frameworks governing this international watercourse are stretched to the limit. A rapid reshaping of the governance of the RGB, from local to state, federal, and international levels, is needed to ensure water security and the continued equitable and fair utilization for both riparian countries. In the face of rapid climate change, it is imperative that decisive actions, informed by careful science, be quickly deployed to ensure the economic vitality of the RGB basin and to limit international conflict over the shared resource.

This case study presents the current state of international governance of the RGB and make policy suggestions centering on: (1) increasing participatory governance and stakeholder engagement and trust; (2) establishing a new mechanism for binational basin management; (3) a new umbrella mechanism to facilitate scientific research in the basin to address surface-groundwater uncertainty and ensure IBWC policy in the region is prepared to handle future climate change and increases in drought.

BACKGROUND

The RGB drains a 330,000-square-mile basin shared between Mexico and the United States and forms the border between the two countries for the last 1,245 miles of its run into the Gulf of Mexico (USIBWC, 2024). For management and hydrological purposes, the RGB has two other main tributaries. The Rio Conchos flows through the Mexican state of Chihuahua before

joining the RGB at the border city of Ciudad Constitución, Mexico. The Pecos River flows through the U.S. states of New Mexico and Texas and joins the RGB at the Amistad Reservoir. The RGB flows 1900 miles from its headwaters in the San Juan Mountains in Southern Colorado to the Gulf of Mexico. The unique hydrology and management of the RGB result in a distinct upper and lower basin, divided below Fort Quitman, TX (Rister et al., 2011). Hydrology in the upper reaches of the RGB is primarily driven by snowmelt runoff. Below Fort Quitman, consumptive use in the upper basin causes the river's discharge to be negligible and sometimes stops flowing altogether. Recharge in the lower basin primarily comes from the Pecos and Rio Conchos rivers, as well as from agricultural and municipal returns (USIBWC, 2024).

Because of the nature of its tributaries and the RGB's role as an international border, both Mexico and the United States have upstream and downstream riparian relationships with each other. Across the RGB basin, average precipitation is low, and most of the basin is arid to semi-arid and has faced worsening extended periods of drought (Rister et al., 2011; USIBWC, 2024). These hydrological and governance features have created a system that faces severe overallocation and complex resource management issues.

International Governance of the RGB is split between two non-overlapping binational agreements that establish a legal framework for allocations and extractions and set quality guidelines (Nava et al., 2016). The 1906 "Convention between the United States and Mexico: Equitable distribution of the waters of the Rio Grande" (the 1906 Convention) establishes a legal framework for the allocation of the international portion of the river between El Paso and Fort Quitman. The Convention requires the U.S. to deliver 60,000 acre-feet/year for Mexican irrigation. Critically, the 1906 Convention allows for the "proportional reduction" of deliveries to

Mexico in the case of "extraordinary drought," but fails to establish specific triggers or metrics for reduced deliveries. (United States & Mexico, 1906)

The “Treaty between the United States of America and Mexico respecting the utilization of waters of the Colorado and Tijuana Rivers and of the Rio Grande” (known as the 1944 Water Treaty) builds on the 1906 Convention by creating a holistic legal framework and governance mechanism for the three major international rivers. The 1944 Water Treaty allocates 1/3 of the water entering the RBG from Mexico to the U.S. and sets a minimum annual delivery of 350,000 acre-feet/year assessed over 5-year cycles. The Treaty also established the International Boundary Water Commission (IBWC) specifically to manage international water resources. The 1944 Water Treaty is endowed with significant flexibility through the Minute Process, which allows the IBWC to update the Treaty to address challenges not initially covered. (United States & Mexico, 1944)

Beginning during the 1992-1997 delivery cycle, Mexico began to default on minimum deliveries to the U.S. As specified in the 1944 Water Treaty, these debts carried over to the next cycle. However, another failure in the 1997-2002 cycle left Mexico with a deficit of 1,328,846 acre-feet (Robb, 2022). It was not until 2005 that Mexico repaid the water debt. However, heavy monsoon rains largely facilitated repayment, and the resolution of debt was mainly due to climatic factors (Rister et al., 2011). A further underdelivery during the 2010-2015 period was cleared in 2016, and in 2021, Mexico narrowly avoided a deficit by transferring all its stored water in the Amistad and Falcon reservoirs (Bueno & Eckstein, 2022). The IBWC used the 1944 Water Treaty minute process to successfully negotiate deficit repayments, yet it has failed to facilitate permanent solutions to address increased drought and regional growth, preferring ad hoc solutions (Garrick et al., 2016). Failures of binational agreements under the Water Treaty

have led to a burden shifting from the binational to the Mexican national level. In turn, the federal-level institutions have shifted the economic and social burden of water deficits in the Rio Conchos basin to local agricultural communities.

The basin faces significant challenges related to RGB management. The 1944 Treaty fails to address surface-groundwater interactions, and national-level frameworks like the Rio Grande Compact (RGC) also lack legal structures for aquifers. Surface-groundwater interactions are a major source of conflict in the region, particularly within the RGC (De Stefano et al., 2018). Consistent overallocation has led to Mexico's treaty noncompliance, further straining the already tense diplomatic relationship. Much of the burden of Mexican water deliveries has fallen on the residents of Rio Conchos, who rely on irrigated agriculture to subsist. Water quality in the basin is a persistent issue, with agricultural and municipal runoff contributing to salinization and pollution (USIBWC, 2024). Compounded by invasive species and the lack of environmental flow allocations, the RGB ecosystem is extremely degraded and at risk (USIWBC, 2024). These challenges are further compounded by climate change, which is already affecting water security in the region and is predicted to cause more frequent and severe droughts (Sandoval-Solis et al., 2002).

Analysis

The 1944 Water Treaty, and specifically the IBWC, facilitates cooperation in the RGB basin. While the Treaty provides an overarching structure, governance of the RGB is a patchwork of overlapping state, federal, and municipal institutions. Although a generalization, the U.S. implements a more decentralized system, emphasizing interstate compacts and local governance, while Mexico relies more heavily on a centralized federal institution: the National Water Agency (CONAGUA) (Garrick et al., 2016; Wilder, 2010). The Mexican National Water

Law (NWL), passed in 1992, theoretically decentralizes governance by creating local management institutions such as river basin councils. However, many of the promised reforms have not been realized (Garrick et al., 2016). In the United States, the RGC and Pecos River Compact (PRC) create institutions to allocate water after meeting the 1906 Conventions deliveries. Within states, irrigation districts and municipal utilities manage local deliveries. Within the RBG framework, the 1906 Convention and the 1944 Water Treaty establish each country's riparian responsibilities toward the other. Increased drought severity and rapid development are straining binational frameworks, and the responsibilities they impose are often burdensome for RGB basin stakeholders. The state and local frameworks are responsible for addressing water shortages caused by mandated deliveries, and the burden is often shifted to stakeholders. Many Mexican RGB stakeholders express dissatisfaction, feel their way of life is under threat, and believe that international interests are being put ahead of domestic use (Robb, 2022).

The effectiveness of RGB governance can be examined through: (1) conflict prevention/resolution, (2) treaty compliance, (3) Stakeholder engagement and perceived legitimacy, (4) adaptive capacity— particularly in the face of climate change, and (5) environmental sustainability.

Institutionally, the 1944 Water Treaty has been only partially effective. Climate change and rapid regional development are challenging the Treaty. While endowed with adaptive capacity through the minute process, passing meaningful minutes can be slow. Recent success with Minutes 319 and 323 on the Colorado River shows that the minute process remains viable, yet the RGB's adaptive capacity is still limited, and the IBWC lacks stakeholder trust (Robb, 2022). The IBWC has successfully negotiated water deficits, but chronic underdeliveries since

1992 clearly show how socio-environmental factors in Mexico conflict with the Treaty obligations. Legally, the Treaty fails to address surface-groundwater interactions, although this issue is receiving more attention recently. The Groundwater question has been a central area of dispute in the RGC since the 1960s, and the failure of the RGC and IBWC to effectively integrate surface-groundwater management into existing frameworks demonstrates a lack of adaptive capacity. Stakeholder engagement with the IBWC is low (Robb, 2022), and Mexico particularly struggles with local engagement, preferring centralized decision-making through CONAGUA (Garrick et al., 2016).

Environmental sustainability within the 1944 Water Treaty, CONAGUA, and the RGC is at best moderately effective. Aquifers are overutilized and depleting across the basin (De Stefano et al., 2018). Water quality, specifically salinization from agricultural return flow, is a persistent problem, as are invasive species and native habitat degradation (Rister et al., 2011). Despite evidence that environmental flows improve overall water availability, implementation in the basin has been limited (Sandoval-Solis et al., 2022).

From the United States' perspective, the most significant failures of RGB governance have been the continued Mexican deficits on the Rio Conchos and the continued reliance on the ambiguous "extraordinary drought" provision of the Treaty, which justifies their noncompliance (Robb, 2022). However, this argument fails to address systemic asymmetries present in the 1944 treaty. Of particular concern, the 350,000 acre-feet delivery on the Conchos needs to be analyzed alongside the 1.5 million acre-feet delivered to Mexico by the Colorado River. Water deliveries on the Colorado are of little consequence to Mexican RGB basin residents due to geography and CONAGUA's failure to allocate benefits. This asymmetry is exacerbated by the 1906 Convention, which allows the U.S. to scale back water deliveries proportionally (the U.S. has

reduced deliveries to Mexico in 30% of the years since 1939 (Garrick et al., 2016)), yet provides no baseline delivery requirements (United States & Mexico 1906) in sharp contrast to Mexican obligations under the 1944 Treaty. Current legal and institutional frameworks fail to address these issues because they operate at scales that encompass the whole border region and use convenient spatial scales to hide inequity (i.e., frame water scarcity in the Rio Conchos as a failure of Mexican governance).

In the face of climate change, drought, and development, the success of the 1944 Water Treaty has relied primarily on ad hoc decision-making. Using our effectiveness criteria, we see the Treaty has failed to meaningfully adapt to changing hydrological and social conditions in the region. Treaty compliance is poor, as shown by continued deficit cycles. While the IBWC has prevented direct conflict and violence, conflict prevention is only moderately effective, as diplomatic water crises are commonplace. Stakeholder engagement is poor, especially in Mexico, where federal compliance continually burdens local communities. The RGB's environmental protection is also poor, with environmental crises like desalination threatening stakeholders and the ecosystem. In the face of these failures, the adaptive capacity of governance, although moderately effective thanks to the minute system, cannot keep up with the rate of change.

Recommendations

The first issues to be addressed are poor stakeholder engagement/lack of trust, systemic power asymmetries, and inequities in the RGB —each of which is a major impediment to the adaptive capacity of existing governance. Addressing these issues is critical for the fair and equitable usage of RGB water, and only through a collaborative, ground-up approach can vitality and economic growth continue as water scarcity increases. Using the minute process, I propose

establishing a new working group whose primary goal is to foster participatory governance within the basin. This group would be specifically tasked with: (1) increasing stakeholder trust and engagement, (2) analyzing federal-level water policy and treaty compliance impacts at the local level. The working group would be composed of members representing national frameworks (e.g., Rio Grande Compact, state irrigation districts, CONAGUA, Mexican river basin councils), NGOs representing different stakeholders and environmental goals, and a binational citizens council representative of RBG users. This group would be responsible for overseeing IBWC initiatives in the RGB basin to ensure participatory governance. Specifically, actions such as annual meetings and the systematic collection of stakeholder input can facilitate increased trust and cooperative management when incorporated into existing IBWC practices.

A shift away from ad hoc decision-making in response to climate-induced crises will be critical to the success of international RGB governance. Already, a changing climate has increased international conflict within the 1944 Water Treaty framework. Without a serious change, it is unlikely the outdated mechanism in place on the RGB will survive. Minutes 319 and 323, which established a framework for binational cooperative basin management on the Colorado River (Robb, 2022), provide an outline for a path forward. The RGB needs a similar minute added to allow for greater binational collaboration. Further, binational basin management can begin to address the asymmetries present in the 1944 Water Treaty deliveries by reducing the unit of management from a three-river system to a basin system. This minute would create a mechanism to promote basin benefit sharing through increased information sharing among practitioners and managers, U.S.-financed infrastructure development, and cooperative water management to promote environmental flows and efficient use. Sharing data and conservation practices at the local level can increase overall water security, and information exchange between

U.S. irrigation districts and Mexican basin organizations would help increase sustainable agricultural practices. Infrastructure funding from Texas irrigation districts and, potentially, already allocated U.S. federal funds through the Infrastructure Bill can fund storage and water-quality infrastructure, which in turn can be used to issue water credits to Mexico for overdeliveries during wet cycles.

Finally, to further shift away from ad hoc decision-making in response to climate change, I recommend the creation of a new research working group tasked explicitly with fostering research related to climate-induced changes in surface-groundwater quantity in the RGB basin. This group would act as an umbrella incorporating existing IBWCC groups like the Transboundary Aquifer Assessment Program and creating new research initiatives to address knowledge gaps. The group would also be tasked with ranking all current and future RGB initiatives, regardless of institution, using a 5-, 10-, and 50-year outlook score for resistance to climate change and drought. Existing projects can be grandfathered in, but future projects must meet a baseline score at each time frame to receive IBWC approval.

While by no means an exhaustive list, these policy suggestions provide a starting point. They are necessarily broad to maintain adaptive capacity and address an uncertain future in the region. These recommendations serve as a baseline to strengthen cooperation and governance and aim to begin creating forward-looking policy that will address climate change. Success in the region will be dictated by scientifically informed policy and a united basin that recognizes the vitality of the RGB as critically linked across international borders.

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